

AI-Powered Hospital Readmission Prevention Program

AI Governance, Ethics & Long-Term Monitoring Plan Version 1.0
Healthcare AI Implementation Portfolio Project

7.1 AI Governance Framework

The purpose of AI governance is to establish accountability, oversight, and safe management of the AI solution throughout its lifecycle.

AI Governance Committee Members:

- Chief Medical Officer
- Chief Information Officer
- Data Science Lead
- Clinical Informatics Team
- Nursing Leadership
- Compliance Officer
- Ethics Representative
- Quality Improvement Team
- Patient Representative

Responsibilities include approving AI use cases, reviewing performance, monitoring safety, and managing updates.

7.2 Ethical AI Principles

Fairness:

The model must provide equitable performance across patient populations and avoid disadvantaging vulnerable groups.

Transparency:

Clinicians should understand important factors influencing AI predictions.

Accountability:

Clear ownership must exist for approval, monitoring, and response to AI-related issues.

Privacy:

Patient health information must be protected through appropriate security and governance practices.

7.3 Continuous AI Performance Monitoring

The AI system will be continuously evaluated after deployment.

Monitoring areas include:

- Accuracy
- Precision
- Recall and sensitivity
- False-positive rates
- False-negative rates
- Clinical outcomes

7.4 Model Drift Monitoring

Healthcare environments change over time. The organization will monitor whether changes in patient populations, clinical practices, or documentation affect model performance.

If performance declines, the model may require retraining and revalidation.

7.5 Bias Monitoring After Deployment

The organization will regularly evaluate AI performance across demographic groups including race, ethnicity, gender, age, insurance status, and socioeconomic factors.

Any significant differences in performance will require investigation and corrective action.

7.6 AI Incident Management

If an AI-related safety concern occurs, the organization will:

1. Document the incident
2. Review the AI recommendation and workflow
3. Identify contributing factors
4. Correct issues
5. Document lessons learned

7.7 AI Lifecycle Management

The AI lifecycle includes:

Development -> Validation -> Deployment -> Monitoring -> Retraining -> Revalidation

Continuous management ensures the AI solution remains safe, accurate, and clinically valuable.

7.8 Long-Term Success Metrics

Patient Outcomes:

- Reduced readmissions
- Improved follow-up care

Safety:

- Reduced missed high-risk patients
- Reduced unnecessary alerts

Equity:

- Similar performance across patient groups

Operations:

- Provider satisfaction
- Workflow improvement

Document Outcome

This document defines the governance, ethical oversight, monitoring, and accountability framework required to maintain a safe and effective healthcare AI solution.